Stock Market Analysis in SQL

Professional insight report | Six NSE stocks | Jan 2015 - Jul 2018

Purpose. Validate the assignment checkpoints, build 20/50-day moving averages and golden-cross signals entirely in SQL, compare six stocks, detect data-quality traps, and correct corporate-action distortions before interpreting returns.

Core finding: the SQL logic reproduces the assignment signal counts, but raw first-to-last returns are misleading for TCS and Infosys because their series contain approximately 50% mechanical price resets caused by 1:1 bonus issues. After adjustment, TCS changes from -23.8% raw to +52.4%, and Infosys from -30.9% raw to +38.2%.

This report is descriptive analysis of the supplied historical data, not investment advice.

1. Data validation and SQL checkpoints

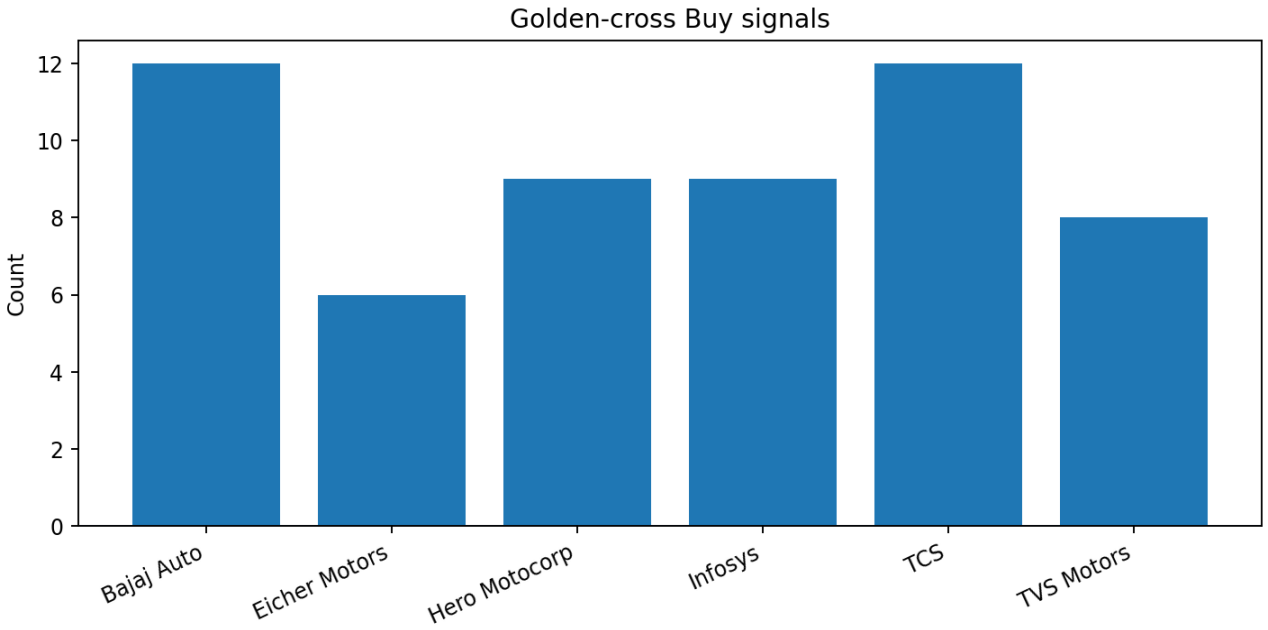
All six CSVs contain 889 rows and the same 13 fields. Dates run from 2015-01-01 to 2018-07-31 after conversion from strings such as 31-July-2018. Six deliverable-quantity NULLs occur across only two dates (2015-12-09 and 2017-08-31), suggesting a reporting-level issue rather than six independent company events.

Checkpoint	Verified result
Bajaj history	889 days | 2015-01-01 to 2018-07-31
Eicher top close	■32,786.40 on 2017-09-07
TCS 2016 average	■2,419.00
Bajaj first MA20	■2,415.53 on 2015-01-29
Bajaj first MA50	■2,283.80 on 2015-03-13
Bajaj 2018-07-31 MA20	■2,918.51
Bajaj signals	12 Buy | 11 Sell | 866 Hold

The moving-average implementation explicitly suppresses partial windows: the 20-day value starts only on row 20 and the 50-day value on row 50. Signals fire only on a crossover day using LAG(), not on every day one average remains above the other.

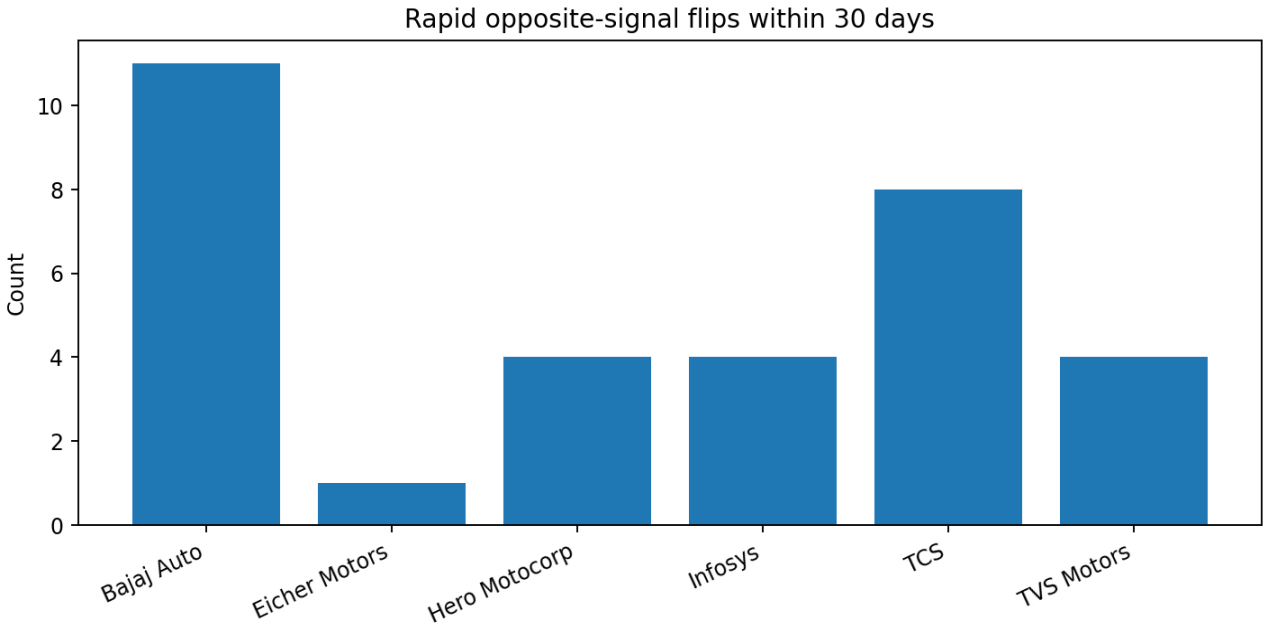
2. Golden-cross signal summary

Stock	Buys	Sells	Latest signal date	Latest signal
Bajaj Auto	12	11	2018-06-21	Buy
Eicher Motors	6	7	2018-06-06	Sell
Hero Motocorp	9	9	2018-05-22	Sell
Infosys	9	9	2018-05-07	Buy
TCS	12	13	2018-06-05	Sell
TVS Motors	8	8	2018-05-17	Sell

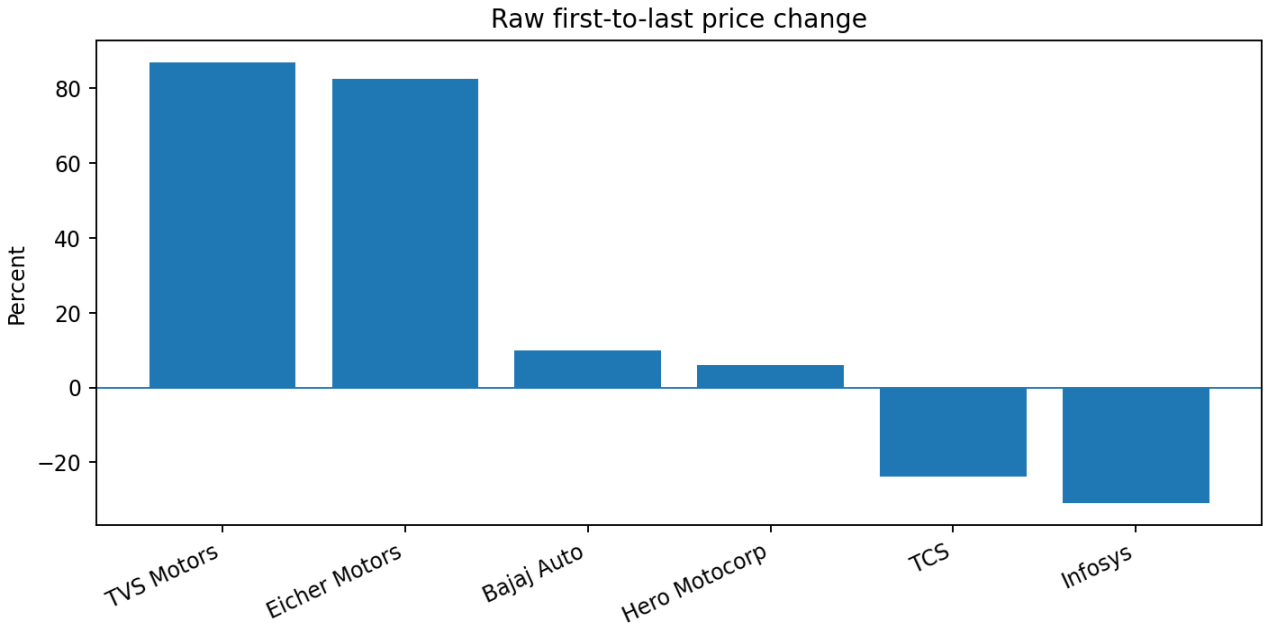


Across all six raw series there are 56 Buy and 57 Sell crossovers, matching the guide checkpoint. A crossover is a lagging trend signal: it can only occur after enough price history exists and after the shorter average has moved through the longer average.

Rapid flips are common in sideways markets. Counting consecutive opposite signals no more than 30 calendar days apart gives: Bajaj 11, TCS 8, Hero 4, Infosys 4, TVS 4, and Eicher 1. Such whipsaws would increase turnover and therefore real-world brokerage, taxes, spread and slippage costs - none of which are present in the dataset.


Rapid opposite-signal flips within 30 days
Count
0
2
4
6
8
10
Bajaj Auto
Eicher Motors
Hero Motocorp
Infosys
TCS
TVS Motors

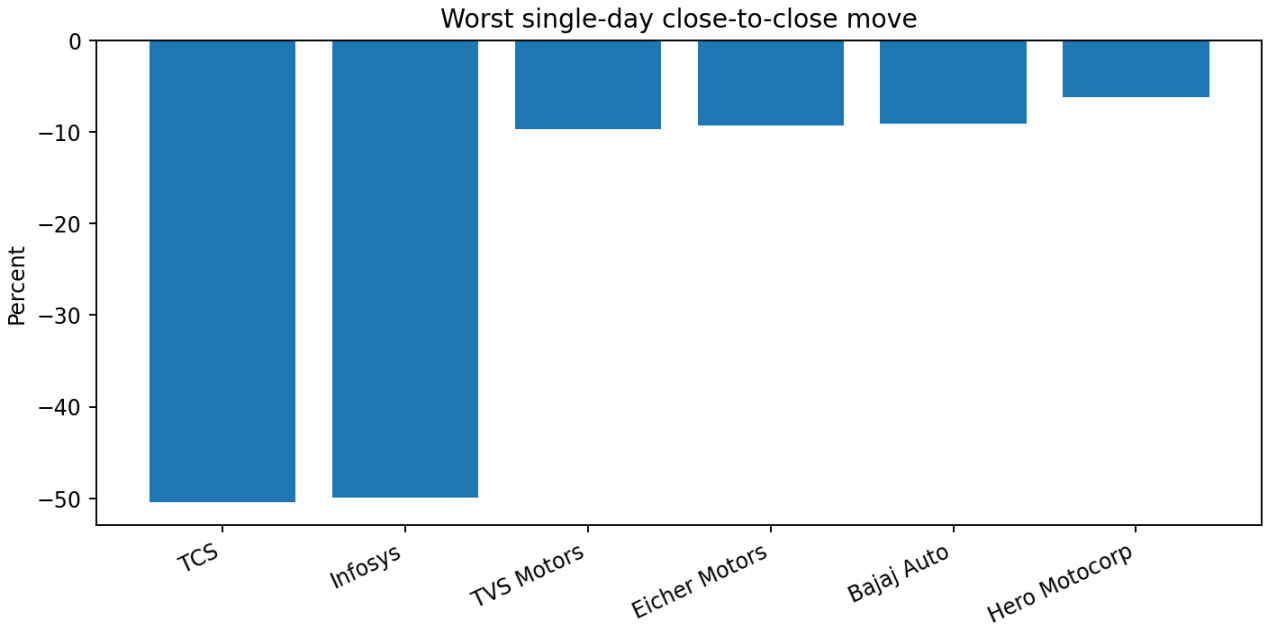
3. Raw returns: useful, but not yet trustworthy



Stock	First close	Last close	Raw change
TVS Motors	■276.85	■517.45	86.9%
Eicher Motors	■15,239.15	■27,820.95	82.6%
Bajaj Auto	■2,454.10	■2,700.70	10.0%
Hero Motocorp	■3,107.30	■3,293.80	6.0%
TCS	■2,548.20	■1,941.25	-23.8%
Infosys	■1,975.80	■1,365.00	-30.9%

The raw ranking puts TVS and Eicher at the top and makes TCS and Infosys look negative over the full period. Before treating that as economic performance, the daily-return diagnostic must be checked for discontinuities.

4. The data trap: corporate actions



Stock	Worst date	Close	Day move
TCS	2018-05-31	■1,744.80	-50.4%
Infosys	2015-06-15	■991.10	-49.9%
TVS Motors	2016-05-03	■287.90	-9.7%
Eicher Motors	2015-08-24	■18,571.75	-9.3%
Bajaj Auto	2015-08-24	■2,188.45	-9.1%
Hero Motocorp	2018-07-23	■3,163.90	-6.2%

TCS and Infosys are obvious outliers at roughly -50% in one day. The supplied guide specifically asks the analyst to investigate these rows rather than accept them as normal losses. TCS identifies a 1:1 bonus with an ex-date of 2018-05-31; Infosys reports a 1:1 bonus issue in 2015. A 1:1 bonus approximately doubles the share count and mechanically halves the per-share price, so pre-event prices must be divided by 2 for an apples-to-apples series.

Adjustment used: TCS dates before 2018-05-31 ÷ 2; Infosys dates before 2015-06-15 ÷ 2. Event-day and later prices remain unchanged.

Stock	Raw full-period change	Adjusted full-period change
TCS	-23.8%	+52.4%
Infosys	-30.9%	+38.2%

This is the most important analytical correction in the project. It overturns the raw conclusion that both technology stocks lost value over the full period.

5. What changes after adjustment?

Rebuilding the moving-average signals on adjusted prices changes TCS from 12 Buys / 13 Sells with a latest Sell on 2018-06-05 to 12 Buys / 12 Sells with a latest Buy on 2018-04-20. That means one raw TCS Sell was created by the corporate-action price cliff rather than by an economic trend reversal. Infosys changes from 9 Buys / 9 Sells to 10 Buys / 10 Sells; its latest signal remains Buy on 2018-05-07.

Therefore the unadjusted signal table is valuable for reproducing the coursework checkpoint, while the adjusted table is the more defensible analytical view for interpretation. The report should show both and explain why they differ.

For Eicher, TVS, Bajaj and Hero, the supplied data does not show the same extreme corporate-action discontinuity in the worst-day diagnostic. Their full-period raw changes are therefore less obviously distorted by this specific issue, though a production investment study would still verify all splits, bonuses, dividends and other corporate actions.

Latest signal vs long-run direction is not a contradiction: a golden cross describes recent moving-average structure, while full-period return summarizes the entire multi-year interval. They answer different questions and operate on different horizons.

6. Interpretation, limitations and submission narrative

Claim - evidence - caveat framework. Every insight in the final presentation should identify the SQL query behind it, state the result, and then state what could invalidate or limit the conclusion. This is especially important for signals because they are computed from historical close prices only.

Method limitations: moving averages are lagging; no dividends are included; no benchmark index is included; no news/fundamentals are included; no position sizing or risk controls are modeled; and transaction costs are absent. The signal counts therefore measure pattern frequency, not realized portfolio profitability.

Professional submission story: (1) prove data coverage and cleanliness, (2) demonstrate correct window-function logic, (3) reproduce the course signal counts, (4) compare raw long-run returns, (5) challenge suspicious results with the worst-day query, (6) adjust TCS and Infosys corporate actions, and (7) explain how the adjustment changes conclusions and signals.

Recommended demo: Use the Streamlit playground to execute the SQL live. Start with the coverage query, then the moving-average query, raw returns, worst-day diagnostic, and adjusted-return query. Use a line chart for moving averages and bar charts for cross-stock comparisons.

Reference basis

Primary project specification: uploaded “Stock market analysis in SQL: student guide” and course deck. Corporate-action verification: TCS investor-relations documentation for the 2018 1:1 bonus/ex-date and Infosys annual-report / investor FAQ documentation for the 2015 1:1 bonus.